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Global GPU Network (GGN): Performance Benchmarking and Comparative Analysis Against Traditional GPU Computing Models

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Abstract

The rapid advancement of artificial intelligence and the booming cryptocurrency market have triggered an unprecedented demand for GPUs, resulting in significant price inflation and limited accessibility to these crucial computational resources. This paper proposes the development of a decentralised Global GPU Network (GGN) by harnessing the underutilised GPU power of mobile devices, offering a cost-effective and efficient alternative to traditional GPU computing models. By leveraging the vast, idle computational resources of smartphones and tablets, the GGN aims to democratise access to GPU power, making high-performance computing more accessible to a broader audience. Through extensive benchmarking and performance analysis, we evaluate the feasibility and effectiveness of the GGN compared to conventional GPU solutions, demonstrating significant cost reductions while maintaining competitive performance levels. Additionally, the GGN model addresses current disparities in revenue distribution within the digital ecosystem, promoting a more equitable and sustainable framework for computational resource allocation. This paper discusses the technical challenges, potential applications, and economic implications of deploying the GGN, highlighting its potential to revolutionise the landscape of GPU computing.

Introduction

The Artificial Intelligence (AI) and cryptocurrency boom has led to an unprecedented increase in GPU (Graphics Processing Unit) demand, causing prices to inflate by as much as 300% (Strzala, 2021). Due to GPUs' unique ability to perform tensor operations and parallel processing, they will play an integral role in our future, especially in a world where AI and blockchain technology serve as fundamental building blocks. Consequently, we should not expect GPU prices to decrease; in fact, further increases seem imminent.

Meanwhile, the processing power in our phones often goes underutilised. Currently, phones are primarily used for social media and messaging applications (Li et al., 2022). These applications are far less computationally intensive than gaming, Virtual Reality (VR), Augmented Reality (AR), and AI applications. Most chip designers optimise phone processors for these less demanding use cases, leaving significant computational power unused.

Another challenge in the digital world is the unequal distribution of funds between content creators and platforms. The revenue-sharing structure of Google AdSense, which provides a 68% revenue share to website publishers, has been controversial due to perceived inequitable fund distribution. While publishers receive 68% of ad revenue, Google retains 32% as a service fee, which many publishers consider excessive, especially considering their operational costs and the value they bring through traffic generation. The recent transition from a CPC (cost-per-click) to a CPM (cost-per-thousand impressions) model has introduced additional complexities. Publishers who previously benefited from high click-through rates may see reduced returns, raising concerns about AdSense's share structure and payment model transparency (Varatharajan, 2024).

These seemingly unrelated problems may have a novel solution: The Global GPU Network (GGN). GGN is a network of mobile phones that harnesses their unused computational power to perform GPU-intensive tasks. The network compensates app developers who integrate its API, allowing them to utilise their users' GPUs for processing. This approach could reduce creators' reliance on advertisement revenue, potentially leading to fewer advertisements across the internet. Beyond being a safe haven for app developers and GPU-focused developers, GGN operates in a decentralised manner without requiring a central server. The network primarily utilizes two advanced technologies: federated learning and layer-2 blockchains. GGN aims to improve revenue for app developers and reduce GPU prices, making GPUs more accessible and democratised, thereby lowering entry barriers and reducing the risk of monopolistic markets in critical technological fields such as big data, AI, visual effects, and more (Shah, 2024).

This research endeavour conducts thorough investigations to verify whether GGN's proposed benefits are realised by its users. We examine this by simulating a real GGN network and benchmarking it on cost, revenue generation, and time delays. While the simulation may differ from a real network in certain aspects, the research aims to provide an accurate representation of actual performance. These benchmarks are then compared with physical GPUs and cloud GPUs. The paper also provides a detailed evaluation that explores distribution provides requisite information for parallel nodes to initiate GPU utilisation. To illustrate, in artificial intelligence applications focused on image recognition, client nodes distribute model parameters and preprocessing data, thereby enabling distributed computational processing where individual nodes contribute GPU resources towards efficient image analysis and classification.

Server nodes, conversely, exist in multiple instances (n), where increased quantity correlates positively with processing velocity, network congestion reduction, and systemic stability. These nodes, typically mobile devices, process specific data packets received from the client node.

The server node operational framework encompasses a sophisticated multi-stage process ensuring both secure data processing and equitable developer remuneration. Initially, data packets undergo processing, followed by smart contract generation. These contracts are implemented on layer-2 blockchain protocols within established networks, exemplified by Polygon on the Ethereum network. This implementation effectively addresses the Blockchain trilemma through expedited processing times and minimal gas fees whilst maintaining parent chain security protocols.

In instances where server nodes reach maximum operational capacity—defined not by computational limits but by optimal performance thresholds—an automated offloading mechanism activates. Each node maintains a distributed hash table (DHT), GGN's future prospects, challenges, and potential solutions to overcome these challenges.

Technical Description About Global GPU Network

To fully comprehend the experimental methodology, it is essential to examine the technical architecture of the Global GPU Network (GGN) and its operational framework, as detailed in (Shah 2024)’s seminal

white paper.

The GGN architecture comprises a dual-node system: server nodes and client nodes. The client node functions as the primary distributor, disseminating target information via shared data packets across the network infrastructure. This comprising identification and location parameters, building upon the foundational work of (Maymounkov & Mazières, 2002). Upon reaching capacity thresholds, the system utilises the DHT to identify proximate nodes with available capacity, continuing this process iteratively until resource availability is confirmed.

The network employs a dynamic pricing mechanism to regulate server node participation, adjusting compensation rates based on network demand and computational availability. During high-demand periods, the system increases compensation rates to attract additional nodes, whilst reducing rates during periods of optimal capacity. This equilibrium is governed by:

$$C_t = C_b * (1 + \alpha \, (\frac{D_t}{S_t} - \beta \,))$$

Where C_t is the current compensation rate, C_b is the base rate, D_t is network demand, S_t is available capacity, α is price sensitivity, and β represents optimal demand-to-supply ratio. A minimum compensation threshold ensures sustainable network participation during low-demand periods.

The network's decentralised architecture ensures robust performance and minimised maintenance requirements. Network expansion occurs through a self-propagating mechanism: new server nodes broadcast their presence to neighbouring nodes, facilitating automatic table updates and information exchange, thereby eliminating centralised server dependencies.

Security protocols encompass comprehensive end-

Figure 2.1

to-end encryption, including encrypted data storage within individual server nodes. The system acknowledges potential security vulnerabilities from both server and client nodes, particularly concerning malware introduction. Mitigation strategies include the implementation of a Software Development Kit (SDK) for clients, ensuring secure data packet transmission and network offloading. Whilst these SDKs demand substantial computational resources—though notably less than model training requirements—they enhance security through process sharding and rigorous verification protocols.

Benefits of the Global GPU Network

The current hypothesis is that the Global GPU Network would be more advantageous for app developers and less expensive than renting a cloud GPU. This conclusion emerged due to the monopolistic nature of both markets. Google has faced multiple lawsuits due to its monopolistic status in advertising (Rogers,

2024), and Nvidia controls over 90% of the GPU market (Moorhead, 2024). Not only are these companies large monopolies, but they often engage in price gouging with their products. For instance, Google takes up to 49% of publishers' revenue in multiple cases, and Nvidia's premium GPU line, H100, has a 1000% profit margin. Both these examples highlight the unfair nature of these markets and why the Global GPU Network would be able to break the controlling monopolistic nature of Nvidia and Google.

It is reasonable to assume that processing time on a Global GPU Network may be significantly longer than using cloud GPUs due to factors like Wi-Fi speed, fluctuations in bandwidth, and the time required to create smart contracts. We can estimate an approximate **20% increase in processing time** when using a global network. However, this additional time could be mitigated if the network is large enough and experiences a high adoption rate, which may lead to improvements in efficiency and reduced overall processing time.

$$H_0 : C_G < C_C$$

$$H_1 : T_G = T_C \times \ (1 + 0.2)$$

C_G : Revenue from using cloud gpu

C_c : Revenue of using GNN

T_G : Time used to train using cloud GPU

T_C : Time used to train using GGN

To test this hypothesis, a simulated network was made, and below are the details about the experiment.

Methodology

This experiment aims to determine if a Global GPU Network (GGN) could provide a more cost-effective solution compared to traditional cloud GPU services while maintaining computational efficiency. The experiment uses time as an independent variable, measuring across different periods (08:00, 14:00, and 00:00) over three days. The dependent variables are the model training duration and associated costs in the GGN environment.

The control experiment utilises a CNN image classification model for Cardiomegaly detection, trained on 1,114 X-ray images over 50 cycles to achieve 80% accuracy. The model contains 7,515,874 parameters. The code was take from: Rahimanshu. (2023, March). Cardiomegaly Disease Prediction Using CNN, Version 1. Retrieved November 3, 2024, from https://www.kaggle.com/datasets/rahimanshu/cardiomegaly-disease-prediction-using-cnn. The control training was conducted on a Tesla T4 GPU through immers.cloud, priced at $0.39 per hour. With a training duration of 10 minutes, the effective cost was $0.0065. For revenue estimation, we used the

average monthly website traffic of 2,000 users (Lindner, 2024) to calculate AdSense revenue in a 10-minute period, resulting in $0.016.

The experimental GGN setup consists of 10 nodes distributed across Ahmedabad's 464.17 km² area. The nodes were strategically placed with an average separation distance of 3.6 kilometers between each node, ensuring optimal network coverage while maintaining reliable connectivity. This spacing was determined through network simulation tests that balanced signal strength and interference patterns.

The implementation code for this experiment is available at github.com/Rehan-shah/ggn-sample-code. The codebase handles data packet distribution, where each packet contains approximately 75MB of data, including model parameters, training instructions, and validation metrics. This packet size was chosen to optimise transmission efficiency while considering typical network bandwidth constraints.

The network utilises iPhone 12 devices, which represent the most common phone model based on device lifecycle analysis (Ewa et al., 2017). Each iPhone 12's Neural Processing Unit (NPU) provides 11 d of computational capacity (Apple, 2020). To maintain sustainable performance levels and prevent thermal throttling, the nodes operate at 30% of maximum capacity. This conservative utilisation ensures consistent performance across extended training periods while preventing device overheating.

Given the small network size and single-buyer scenario, a dynamic pricing mechanism would not effectively represent market forces (Robinson, 1969). Instead, the experiment employs a fixed pricing strategy where costs are calculated directly. This approach enables a thorough cost-benefit analysis focusing on creating mutual advantages for both developers and GPU providers. The analysis examines various revenue-sharing scenarios, seeking an equilibrium point where benefits are equally distributed. For example, if GPU users experience a 50% cost reduction compared to cloud services, the system should generate a 2x increase in developer revenue to maintain balanced incentives. This balanced approach ensures sustainable network participation while maintaining competitive pricing.

Environmental controls were strictly maintained throughout the experiment. WiFi speeds were consistently monitored at 100 ±10 Mbps download and 70 ±6 Mbps upload, with each node's connection stability verified before each training session. To control WiFi speeds, network traffic shaping tools were employed to allocate specific bandwidth limits to each node, ensuring uniform speed distribution and preventing network congestion. Docker containers were implemented to standardise the computing environment across the varying hardware configurations, ensuring computational consistency despite underlying hardware differences.

The BNB Smart Chain processes all transactions, with smart contract costs varying due to blockchain price volatility. The smart contract implementation includes automated payment distribution mechanisms and performance verification protocols. Transaction costs were recorded for each training session without outlier removal, acknowledging the inherent volatility of blockchain transaction fees.

The experiment was structured to simulate real-world conditions while maintaining experimental control. Data collection focused on two primary metrics:

1. The time required to train the model on the simulated network, including data distribution and aggregation periods

2. The cost of creating and executing smart contracts at different time periods, with full acknowledgment of blockchain price volatility

The experiment ran nine times across three daily periods over three days to capture temporal variations in performance and costs. Each session recorded complete training cycles, including network initialisation, data distribution, model training, and result aggregation phases. This comprehensive approach ensures that all aspects of the network's operational characteristics were captured in the experimental data.

Conclusion

The experimental results provided significant insights into the viability of the Global GPU Network, yielding mixed outcomes regarding our initial hypotheses.

The experimental results showed a mean processing time of 19 minutes 43 seconds, significantly exceeding our hypothesised 20% increase over the control's 10-minute baseline. The substantial standard deviation of 4 minutes 34 seconds indicates significant instability in processing times, with a 95% confidence interval ranging from 16 minutes 12 seconds to 23 minutes 13 seconds. This high variance is particularly concerning as it suggests unpredictable completion times for developers. In real-world implementations, this variance could be even more pronounced due to diverse network conditions and varying levels of congestion across different locations.

Analysis of temporal variations revealed notable patterns. Sessions at 08:00 and 00:00 averaged T, while 14:00 sessions consistently showed longer durations. These variations aligned with local network congestion patterns rather than blockchain network activity. According to (Pierro & Rocha, 2019), blockchain congestion typically peaks during UTC market hours. When converting our test times to UTC (03:30, 08:30, and 18:30), they fell outside these peak periods, suggesting that local network conditions, rather than blockchain congestion, were the primary performance bottleneck. This finding is further supported by the higher processing times during Indian peak hours at 14:00, despite this corresponding to relatively quiet blockchain network periods.

Table 5.1

	19th oct	20th oct	21st oct
9:00 am	2x|50%	2x|50%	2x|50%
2:00 pm	2x|40%	2x|40%	2x|40%
12:00 am	2x|25%	2x|25%	2x|25%

The cost analysis demonstrated promising results with an average smart contract deployment cost of $0.00980. The table 5.1, is an infographic which showcase the variance over different times and days. The table 5.1 use multiple|discount format, where the 1st number represent the revenue multiple the developer would enjoyed if he integrated the network and the 2nd variable showcase the amount of discount given to the GPU users if used the network. There is consistence of cost, in respect to experimental accuracy, through the days, the main variations occurs during different timing. Examining the relationship between deployment times and blockchain activity revealed interesting patterns. According to (Pierro & Rocha, 2019), blockchain congestion and gas fees peak during UTC market hours. Our morning deployments at 09:00 IST (03:30 UTC), occurring before market hours, consistently achieved the most favourable conditions, enabling 50% discounts to GPU users while doubling developer revenue. Afternoon sessions at 14:00 IST (08:30 UTC), coinciding with early market hours, showed slightly increased costs but still maintained 40% discounts. The midnight deployments at 00:00 IST (18:30 UTC), falling during market hours, demonstrated the lowest benefits with 25% discounts, though still maintaining doubled developer revenue. Although this regionally significantly benefits the network, globally, crypto congestion times might increasingly coincide with the network's regular operations. Moreover, as the network scales, ensuring continuous 24/7 functionality may lead to additional overlaps with American working hours leading to further price increase.

However, these results come with important caveats. Our testing occurred during relatively stable market conditions. During periods of extreme blockchain congestion, such as major market events or network-wide spikes in activity, gas fees could increase substantially. Such scenarios could temporarily eliminate the cost advantages over traditional GPU solutions, making the network economically unviable during these periods. This potential instability suggests that while the network generally provides significant cost benefits, these advantages may not be consistently available during periods of high blockchain network stress.

Comparing the GGN with the Tesla T4 GPU reveals that while the Tesla T4, priced at $1,532.59, offers impressive performance, it may not always be the most cost-effective option. For long-term, large-scale data processing tasks, purchasing a dedicated GPU like the Tesla T4 can be more beneficial. However, for shorter-term projects, GGN provides a more economical and flexible alternative.

In conclusion, the Global GPU Network shows promising economic viability with consistent 25-50% cost savings and doubled developer revenue, despite challenges in processing time stability. While the current implementation proves cost-effective during off-peak hours, future success depends on addressing processing time variations and maintaining performance during high blockchain congestion periods.

Future Prospects

The Global GPU Network's initial findings open several promising avenues for future research and development. A critical next step involves conducting extensive experiments with networks of 50+ nodes to validate the effectiveness of dynamic pricing mechanisms across different time zones and usage patterns. This expanded testing would provide crucial insights into how price elasticity affects network participation and stability over extended periods. Additionally, the network's potential could be significantly enhanced by expanding device compatibility beyond mobile phones. Integration with WebGPU-enabled devices, IoT systems, and laptops could create a more diverse and robust computational network, though this would require careful study of performance variations across different device types and their impact on overall network reliability.

Scale remains a crucial factor in network effectiveness, necessitating large-scale simulations with 1000+ nodes to determine optimal network size and identify critical mass points where network efficiency surpasses traditional GPU solutions. Such scaling studies would help identify potential bottlenecks and inform infrastructure requirements for widespread deployment. To validate the network's practical utility, primary research involving interviews with potential stakeholders - including app developers currently using cloud GPUs, mobile application developers seeking alternative revenue streams, and AI researchers requiring distributed computing resources - would provide valuable insights into adoption likelihood and desired features.

To address stability concerns and improve network reliability, implementing city-specific private networks as layer-3 chains on the Binance Smart Chain could significantly reduce operational costs while improving network stability. This architectural evolution would involve creating localised processing centres in metropolitan areas, providing more predictable performance by reducing dependency on global network conditions. Such an implementation could serve as a prototype for future urban-focused computational networks, though careful consideration of security implications in multi-layer chain architectures would be essential.

Given current resource constraints, initial implementation will focus on expanding the test network within Ahmedabad to 50 nodes over the next three months. This expanded network will enable thorough testing of dynamic pricing mechanisms and provide more robust performance data. Following successful local expansion, development will progress to implementing WebGPU integration, allowing for a more diverse device pool and increased network capacity. The research findings and implementation experience gained from these initial phases will inform the eventual deployment of city-specific layer-3 chains, with the goal of creating a more stable and efficient computational network.

Declarations

Authors Contribution

R.S. is the sole author of this manuscript and is responsible for all aspects of the research presented. R.S. conceived the idea of the Global GPU Network (GGN), conducted the performance benchmarking and comparative analysis against traditional GPU computing models, wrote the main manuscript text, and prepared all figures and tables. R.S. also reviewed and revised the manuscript to ensure its accuracy and clarity.

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Availability of data and material

Data booklet and calculations are provided in calculations.tex.

Competing interests

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Figures

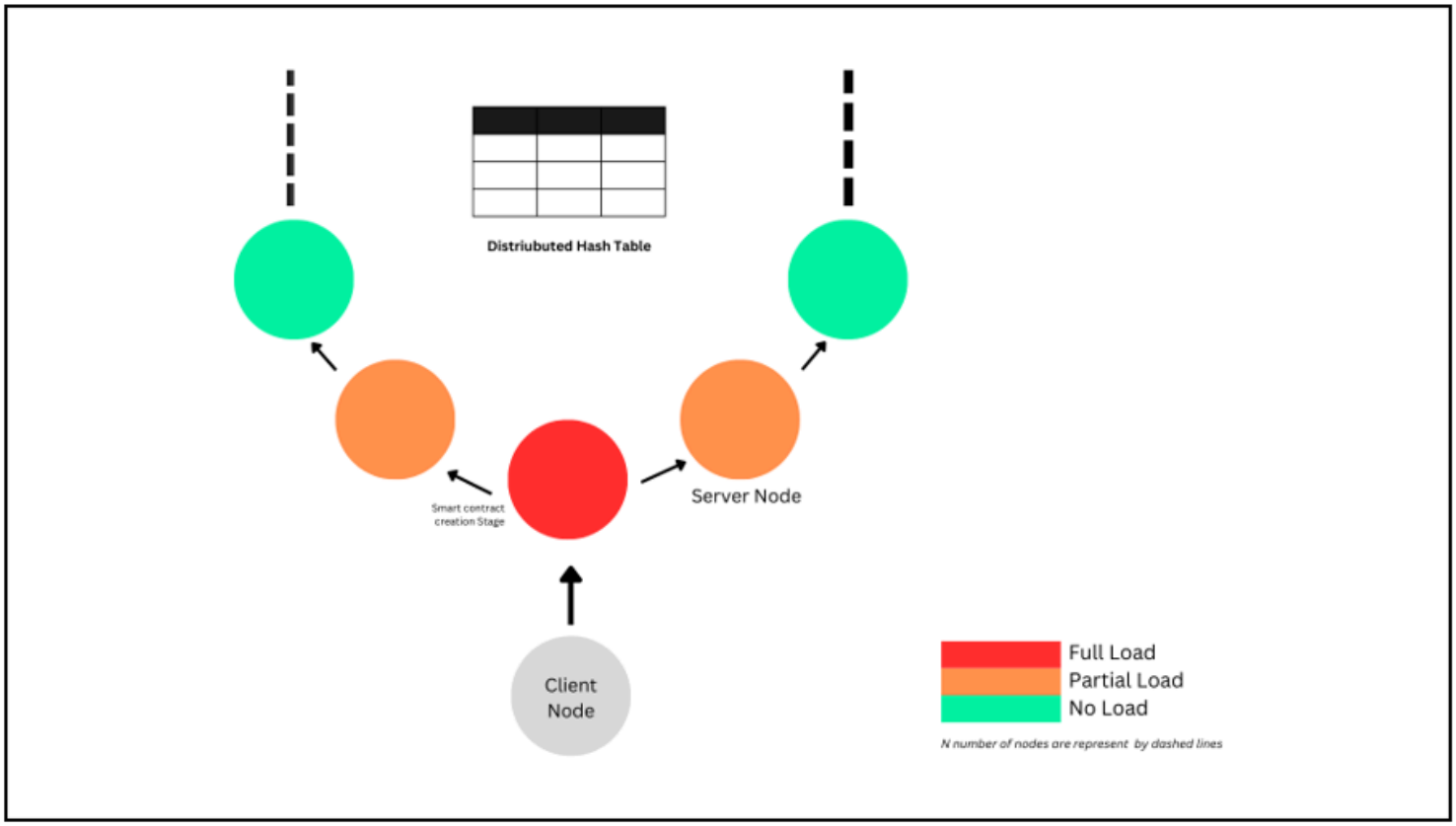

Figure 1

Legend not included with this version

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